

Dana S. Scott Bibliography

Draft of Sunday, March 15, 2026

BOOKS

- a. Scott, D. and Lemmon, E. An Introduction to Modal Logic, (edited by K. Segerberg) American Philosophical Quarterly Monograph Series, No. 11, Oxford University Press, 1977, x + 96 pp.
- b. Scott, D., Gierz, G., Hofmann, K., Keimel, K., Lawson, J., and Mislove, M., A Compendium of Continuous Lattices, Springer-Verlag, 1980, xx + 371 pp.
- c. Scott, D., Gierz, G., Hofmann, K., Keimel, K., Lawson, J., and Mislove, M., Continuous Lattices and Domains, Cambridge University Press, 2003, 600 pp. ISBN: 0521803381 [Revised ed. of b.]
- d. The Univalent Foundations Program. Homotopy Type Theory: Univalent Foundations of Mathematics. The Institute for Advanced Study, Princeton, N.J., 2013, viii + 586 pp.

PAPERS

- 1. Scott, D. and Kalicki, J. Equational completeness of abstract algebras. Koninkl. Nederl. Akademie van Wetenschappen, Proceedings, Series A, vol. 58 (1955), pp. 650 - 659.
- 2. Scott, D. Equationally complete extensions of finite algebras. Koninkl. Nederl. Akademie van Wetenschappen. vol. 59, Proceedings, Series A, 1956, pp. 35 - 38.
- 3. Scott, D. A symmetric primitive notion of Euclidean geometry. Koninkl. Nederl. Akademie van Wetenschappen, Proceedings, Series A, vol. 59 (1956), pp. 456 - 461.
- 4. Scott, D. and Roth, J. A vector method for solving linear equations and inverting matrices. Journal of Mathematics and Physics, vol. 35 (1956), pp. 312 - 317.
- 5. Scott, D. The independence of certain distributive laws in Boolean algebras. Transactions of the American Mathematical Society, vol. 84 (1957), pp. 258 - 261.
- 6. Scott, D. and Tarski, A. The sentential calculus with infinitely long expressions. Colloquium Mathematicum, vol. 6 (1958), pp. 165 - 170.
- 7. Scott, D., and Suppes, P. Foundational aspects of theories of measurement. The Journal of Symbolic Logic, vol. 23 (1958), pp. 113 - 128.
- 8. Scott, D., Gal, I.L., and Rosser, J.B. Generalization of a lemma of G.F. Rose. The Journal of Symbolic Logic, vol. 23 (1958), pp. 137 - 138.
- 9. Scott, D. Dimension in elementary Euclidean geometry. In: The Axiomatic Method, with Special Reference to Geometry and Physics, edited by L. Henkin, et al.. vol. 35, North- Holland Publishing Company, Amsterdam, 1959, pp. 53 - 67.

10. Scott, D. and Rabin, M. Finite automata and their decision problems. IBM Journal of Research and Development, vol. 3 (1959), pp. 114 - 125.
11. Scott, D. On constructing models for arithmetic. In: Proceedings of the International Symposium on Foundations of Mathematics: Infinitistic Methods. Panstwowe Wydawnictwo Naukowe, 1959, pp. 235 - 255.
12. Scott, D. On a theorem of Rabin. Koninkl. Nederl. Akademie van Wetenschappen, Proceedings, Series A, vol. 63 (1960), pp. 481 - 484.
13. Scott, D. More on the axiom of extensionality. In: Essays on the Foundations of Mathematics, edited by Y. Bar-Hillel, E. Poznanski, M. Rabin, and A. Robinson. The Magnes Press, The Hebrew University, 1961, pp. 115 - 131.
14. Scott, D. Measurable cardinals and constructible sets. Bull. of the Polish Academy of Science, vol. 9 (1961), pp. 521 - 524.
15. Scott, D. Quine's individuals. In: Logic, Methodology and Philosophy of Science, Proceedings of the 1960 International Congress, edited by E. Nagel, P. Suppes, and A. Tarski. Stanford University Press, 1962, pp. 111 - 115.
16. Scott, D. Algebras of sets binumerable in complete extension of arithmetic. In: Recursive Function Theory, edited by J. Dekker. American Mathematical Society, 1962, pp. 117 - 121.
17. Scott, D., Frayne, T., and Morel, A. Reduced direct products. Fundamenta Mathematicae, vol. 51 (1962), pp. 195 - 228. Correction, vol. 53 (1963) p.117.
18. Scott, D. and Monk, D. Additions to some results of Erdos and Tarski. Fundamenta Math., vol. 53 (1964), pp. 335 - 343.
19. Scott, D. Measurement structures and linear inequalities. Journal of Mathematical Psychology, vol. 1 (1964), pp. 233 - 247.
20. Scott, D. Invariant Borel sets. Fundamenta Mathematicae, vol. 54 (1964), pp. 117 - 128.
21. Scott, D. Logic with denumerably long formulas and finite strings of quantifiers. In: The Theory of Models, Proceedings of the 1963 International Symposium at Berkeley, edited by J. Addison, et al. North-Holland, Amsterdam, 1965, pp. 329 - 341.
22. Scott, D. and Krauss, P. Assigning probabilities to logical formulas. In: Aspects of Inductive Logic, edited by J. Hintikka and P. Suppes. North Holland Publishing Company, 1966, pp. 219 - 264.
23. Scott, D. Existence and description in formal logic. In: Bertrand Russell: Philosopher of the Century, edited by R. Schoenman. George Allen & Unwin, London, 1967, pp. 181 - 200.

24. Scott, D. A proof of the independence of the continuum hypothesis. *Math. Systems Theory*, vol. 1 (1967), pp. 89 - 111.
25. Scott, D. Some definitional suggestions for automata theory. *J. of Computer and System Sci.*, vol. 1 (1967), pp. 187 - 212.
26. Scott, D. Extending the topological interpretation to intuitionistic analysis. *Compositio Math.*, vol. 20 (1968), pp. 194 - 210.
27. Scott, D. Boolean models and nonstandard analysis. In: *Applications of Model Theory to Algebra, Analysis, and Probability*, edited by W. Luxemburg. Holt, Rinehart and Winston, Inc., 1969, pp. 87-92.
28. Scott, D. On completing ordered fields. In: *Applications of Model Theory to Algebra, Analysis, and Probability*, edited by W. Luxemburg. Holt, Rinehart and Winston, Inc., 1969, pp. 274 - 278.
29. Scott, D. Constructive validity. In: *Symposium on Automatic Demonstration*, edited by M. Landet, D. Lacombe, L. Nolin, and M. Scheutzenberger. Springer-Verlag, LNM 125, 1970, pp. 237 - 275.
30. Scott, D. Advice on modal logic. In: *Philosophical Problems in Logic, Some Recent Developments*, edited by K. Lambert. D. Reidel Publishing Company, 1970, pp. 143 - 173.
31. Scott, D. The problem of giving precise semantics for formal languages. In: *Linguaggi nella Societa e nella Tecnica*,. Edizioni di Comunita, 1970, pp. 225 - 254.
32. Scott, D. Extending the topological interpretation to intuitionistic analysis, II. In: *Proceedings of Conference on Proof Theory and Intuitionism*, edited by J. Myhill. North-Holland Publishing Company, 1970, pp. 235 - 255.
33. Scott, D. Outline of a mathematical theory of computation. In: *Proc. of the Fourth Annual Princeton Conf. on Info. Sciences and Systems*, 1970, pp. 169-176.
34. Scott, D. Semantical archaeology: A parable. *Synthese*, vol. 11 (1970), pp. 150 - 158.
35. Scott, D. The lattice of flow diagrams. In: *Semantics of Algorithmic Languages*, edited by E. Engeler. Springer-Verlag, LNM 188, 1971, pp. 311 - 368.
36. Scott, D. and Myhill, J. Ordinal definability. In: *Axiomatic Set Theory, Proceedings of Symposia in Pure Mathematics*, edited by D. Scott and T. Jech. vol. 13, American Mathematical Society, 1971, pp. 271 - 278.
37. Scott, D. On engendering an illusion of understanding. *The Journal of Philosophy*, vol. 68 (1971), pp. 787 - 807.

38. Scott, D. and Strachey, C. Toward a mathematical semantics for computer languages. In: Proc. of the Symp. on Computers and Automata. Microwave Reserach Institute Symposia Series, vol. 21, Polytechnic Press, Brooklyn N.Y., 1971, pp. 19 - 46.
39. Scott, D. Lattice theory, data types and semantics. In: Formal Semantics of Programming Languages, edited by R. Rustin. Courant Computer Science Symposis, vol. 2, Prentice Hall, 1972, pp. 65 - 106.
40. Scott, D. Continuous lattices. In: Toposes, Algebraic Geometry and Logic, edited by F. Lawvere. Springer-Verlag, LNM 274, 1972, pp. 97-136.
41. Scott, D. Mathematical concepts in programming language semantics. AFIPS Spring Joint Computer Conference, Proceedings, vol. 40 (1972), pp. 225 - 234.
42. Scott, D. Background to formalization. In: Truth, Syntax and Modality, edited by H. Leblanc. North-Holland Publishing Company, 1972, pp. 244-273.
43. Scott, D. Data types as lattices. Slides from lectures at Amsterdam, June 1972, ii + 90 pp.
44. Scott, D. Models for various type-free calculi. In: Logic, Methodology and Philosophy of Science IV, edited by P. Suppes et al. North-Holland Publishing Company, 1973, pp. 157 - 187.
45. Scott, D. Axiomatizing set theory. In: Proceedings of Symposia in Pure Mathematics, edited by T. Jech. vol. 13, part 2, American Mathematical Society, 1974, pp. 207 - 214.
46. Scott, D. Completeness and axiomatizability in many-valued logic. In: Proceedings of the Tarski Symposium, edited by L. Henkin et al. vol. 25, American Mathematical Society, 1974, pp. 411 - 435.
47. Scott, D. Rules and derived rules. In: Logical Theory and Semantic Analysis, edited by S. Stenlund. D. Reidel Publishing Company, 1974, pp. 147 - 161.
48. Scott, D. Lambda-calculus and recursion theorem (Preliminary version). In: Proceedings of the Third Scandinavian Logic Symposium, edited by S. Kanger. North-Holland Publishing Company, 1975, pp. 154 - 193.
49. Scott, D. Combinators and classes. In: Lambda-Calculus and Computer Science Theory, edited by C. Bohm. Springer-Verlag, LNCS 37, Springer-Verlag, 1975, pp. 1 - 26.
50. Scott, D. Some philosophical issues concerning theories of combinators. In: Lambda-Calculus and Computer Science Theory, edited by C. Bohm. Springer-Verlag, LNCS 37, Springer-Verlag, 1975, pp. 346 - 366.
51. Scott, D. Does many-valued logic have any use? In: Philosophy of Logic, edited by S. Korner. Basil Blackwell, Oxford, 1976, pp. 64 - 74.

52. Scott, D. Data types as lattices. *SIAM Journal on Computing*, vol. 5 (1976), pp. 522-587. (Revised version of 43, also distributed as Technical Monograph PRG-5, September 1976, Oxford University Computing Laboratory Programming Research Group.)
53. Scott, D. Logic and programming languages. *Communications of the Association for Computing Machinery*, vol. 20 (1977), pp. 634 - 641.
54. Scott, D. Foreword. In: Stoy, J.E. *Denotational Semantics*. MIT Press, 1977, pp. 1- 16.
55. Scott, D. Foreword. In: Bell, J.L. *Boolean-valued models for independence proofs*. Oxford University Press, 1977, pp. xi - xviii.
56. Scott, D. A note on distributive normal forms. In: *Essays in Honour of Jaakko Hintikka*, edited by E. Saarinen et al. D. Reidel Publishing Company, 1979, pp. 75 - 90.
57. Scott, D. Identity and existence in intuitionistic logic. In: *Applications of Sheaves*, edited by M. Fourman. *Durham Proceedings (1977)*, Springer-Verlag, LNM 753, 1979, pp. 660 - 696.
58. Scott, D. and Fourman, M. Sheaves and logic. In: *Applications of Sheaves, Durham Proceedings (1977)*, edited by M. Fourman. Springer-Verlag, LNM 753, 1979, pp. 302 - 401.
59. Scott, D. Lambda calculus: Some models, some philosophy. In: *The Kleene Symposium*, edited by J. Barwise et al. North-Holland Publishing Company, 1980, pp. 223 - 265.
60. Scott, D. Relating theories of the lambda-calculus. In: *To H.B. Curry: Essays on Combinatory Logic, Lambda Calculus and Formalism*, edited by J. Seldin and J. Hindley. Academic Press, 1980, pp. 403 - 450.
61. Scott, D. Some ordered sets in computer science. In: *Ordered Sets*, edited by I. Rival. D. Reidel Publishing Company, 1982, pp. 677 - 718.
62. Scott, D. Domains for denotational semantics. In: *Automata, Languages and Programming*, edited by M. Nielsen and E. Schmidt. *Ninth Colloquium, Aarhus, Denmark, July 1982 (ICALP 82)*, Springer-Verlag, LNCS 140, 1982, pp. 577 - 610.
63. Scott, D. Lectures on a mathematical theory of computation. In: *Theoretical Foundations of Programming Methodology*, edited by M. Broy and G. Schmidt. D. Reidel Publishing Company, 1982, pp. 145 - 292.
64. Scott, D. Foreword. In: Bloom, S.L. (ed). *Calvin C. Elgot Selected Papers*. Springer-Verlag, 1982, v - ix.
65. Scott, D. and Scherlis, W. First steps towards inferential programming. In: *Information Processing 83, Proceedings of the IFIP 9th World Computer Congress*, edited by R. Mason. IFIP Congress Series, vol. 9, North-Holland Publishing Company, 1983, pp. 199 - 212.

66. Scott, D. and Scherlis, W. L. Semantically Based Programming Tools (Summary). In: Ehrig, H. et al. (eds.), TAPSOFT Berlin, 1985, vol. 1, Springer-Verlag Lecture Notes in Computer Science, pp. 52–59.
67. Scott, D., McCarty, D.C., and Harty, J.F. Bibliography. In: Feferman, S. and Barwise, J. Model-Theoretic Logics, Springer-Verlag, Perspectives in Mathematical Logic, 1985 pp. 793 - 891.
68. Scott, D. What should we demand of electronic dictionaries? In: The Uses of Large Text Databases, 3rd Annual Conference of the UW Centre for the New Oxford English Dictionary, Waterloo, Canada, 1987, pp. 1 - 15.
69. Scott, D. Capturing concepts with data structures. In: Data and Knowledge (DS-2), edited by R. Meersman and A. Sernadas. North-Holland Publishing Company, 1988, pp. 341 - 354.
70. Scott, D. Subject Classification and Natural-Language Processing for Retrieval in Large Databases. In: Richardson, G., Proceedings of Classification Theory in the Computer Age: Conversations Across the Disciplines, 1989, pp. 65 - 74.
71. Gunter, C. A., and Scott, D. S. Semantic Domains and Denotational Semantics. In: Handbook of Theoretical Computer Science, Vol. B (Formal Models and Semantics), edited by J. van Leeuwen, North-Holland, 1990, pp. 633 - 674.
72. Scott, D. Teaching projective geometry. In: MCS News '90, vol. 8 (1990), Mellon College of Science, Carnegie Mellon University, pp. 1 - 5.
73. Scott, D. Existence and description in formal logic. In: Philosophical Application of Free Logic, edited by K. Lambert. Oxford University Press, 1991, pp. 28 - 48. (Revised version of No. 23 with additions.)
74. Scott, D. Exploration with Mathematica. In: Rashid, R., Carnegie Mellon Computer Science: A 25-Year Commemorative, Ch. 22, Addison-Wesley, 1991, pp. 501 - 515.
75. Scott, D. S. Will Logicians be Replaced by Machines? In: TACS 1991, p. 771.
76. Scott, D., Freyd, P. J., Mulry, P., and Rosolini, G. Extensional PERs. Information and Computation, vol. 98 (1992), pp. 211 - 227.
77. Scott, D. S. A Type-Theoretical Alternative to ISWIM, CUCH, OWHY. Theoretical Computer Science, vol. 121 (1993), pp. 411- 440.
78. Scott, D. Symbolic Computation and Teaching. In: Artificial Intelligence and Symbolic Mathematical Computation, AISMC-3, edited by J. Calmet, et al., vol. 1138 (1996), Springer-Verlag Lecture Notes in Computer Science, pp. 1 - 20.
79. Scott, D. S. Effective Versions of Equilogical Spaces. In: Workshop on Domains, Electronic Notes in Theoretical Computer Science 1998, p. 204.

80. Scott, D., Birkedal, L., Carboni, A. Rosolini, G. Type Theory via Exact Categories: Extended Abstract. Proceedings of the 13th Annual IEEE Symposium on Logic in Computer Science, 1998, pp. 188 - 198.
81. Scott, D. Some Reflections on Strachey and his Work. In: A Special Issue Dedicated to Christopher Strachey, edited by O. Danvy and C. Talcott. Higher-Order and Symbolic Computation, vol. 13 (2000), pp. 103 - 114.
82. Scott, D., Awodey, S., Birkedal, L. Local Realizability Toposes and a Modal Logic for Computability. Mathematical Structures in Computer Science, vol. 12 (2002), pp. 319 - 334.
83. Scott, D., Bauer, A., Birkedal, L. Equilogical Spaces. Theor. Comput. Sci., vol 315 (2004), pp. 35 - 59.
84. Scott, D., Foreword. In: The Seventeen Provers of the World, edited by Freek Wiedijk, Birkhäuser, 2006, pp. i - xii.
85. Scott, D., The Algebraic Interpretation of Quantifiers: Intuitionistic and Classical. In: Seventy Years of Foundational Studies: A Tribute to Andrzej Mostowski, edited by A. Ehrenfeucht, et al., IOS Press, 2008. pp. 289 - 312.
86. Scott, D. and McCarty, D. C. A., Reconsidering ordered pairs. Bull. Symbolic Logic, vol. 14 (2008), pp. 379 - 397.
87. Scott, D.. Stochastic λ -calculi: An extended abstract. Journal of Applied Logic , vol. 12 (2014), pp. 369 - 376.
88. Scott, D., Plotkin, G.D., Bauer, A. Cartesian Closed Categories of Separable Scott Domains. Theoretical Computer Science, vol. 546 (2014), pp. 17 - 29.
89. Scott, D. and Benz Müller, C. Automating Free Logic in Isabelle/HOL. In: Mathematical Software (ICMS 2016), Proceedings, G.-M. Greuel, T. Koch, P. Paule, A. Sommese (eds.), Springer, LNCS, volume 9725 (2016), pp. 43 - 50.
90. Scott, D., and Fritz, P., Lederman, H., Liu T., Can Modalities Save Naive Set Theory? The Review of Symbolic Logic, vol. 11 (2018), pp. 21 - 47.
91. Scott, D. and Benz Müller, C. Some Reflections on a Computer-aided Theory Exploration Study in Category Theory (Extended Abstract). In: 3rd Conference on Artificial Intelligence and Theorem Proving (AITP 2018), Thomas C. Hales, Cezary Kaliszyk, Stephan Schulz, Josef Urban (eds.), 2018.
92. Scott, D. and Benz Müller, C. Axiom Systems for Category Theory in Free Logic. Archive of Formal Proofs, 2018, pp. 1 - 12.

93. G. Bacci, R. Furber, D. Kozen, R. Mardare, P. Panangaden, and D. Scott. Boolean-valued semantics for the stochastic λ -calculus. In: Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science, LICS 2018, Oxford, UK, July 09-12, 2018, pages 669 - 678.
94. Tamar Lando and Dana Scott. A calculus of regions respecting both measure and topology. *Journal of Philosophical Logic*, vol. 48 (2019), pp. 825 - 850.
95. Scott, D. and Benzmüller, C. Automating Free Logic in HOL, with an Experimental Application in Category Theory. *Journal of Automated Reasoning*, vol. 64 (2020), pp. 53 - 72. (The associated dataset is published as: Axiom Systems for Category Theory in Free Logic. Archive of Formal Proofs. 2018.)
96. Scott, D., Tiemens, L., Benzmüller, C., and Benda, M. Computer-supported Exploration of a Categorical Axiomatization of Modeloids. In: Proceedings of Relational and Algebraic Methods in Computer Science: 18th International Conference, RAMiCS 2020, Palaiseau, France, April 8–11, 2020, Springer Lecture Notes in Computer Science, vol. 12062 (2020), pp. 302 - 317.
97. Scott, D., Bayer, J., Gonus, A., and Benzmüller, C. Category Theory in Isabelle/HOL as a Basis for Meta-logical Investigation. In: Dubois, C., & Kerber, M., editor(s), Proceedings of Intelligent Computer Mathematics - International Conference, CICM 2023, September 4-8, 2023, Cambridge, UK. Springer Lecture Notes in Computer Science, vol. 14101 (2023), pp. 69 - 83.
98. Scott, D. and Benzmüller, C. Notes on Gödel's and Scott's Variants of the Ontological Argument. *Monatshefte für Mathematik*, vol. 208 (2025), pp. 569 - 611. (The associated dataset is published as: Notes on Gödel's and Scott's Variants of the Ontological Argument.(Isabelle/HOL dataset). Archive of Formal Proofs. 2025.)
99. Scott, D., Furber, R., Mardare, R., and Panangaden, P. Interpreting Lambda Calculus in Domain-Valued Random Variables. In: 34th EACSL Annual Conference on Computer Science Logic (CSL 2026), Editors: Stefano Guerrini and Barbara König, pp.48:1–48:20.